

Deep Learning

Theoretical Exercises - Lesson 4 - Chapter 5

Exercises on the book "Deep Learning" written by Ian Goodfellow,
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Exercises and solutions by T. Méndez and G. Schuster

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1 Exercises on Machine Learning Basics



Hint:

Several answers are correct in the multiple choice exercises.

1. The goal of machine learning is to achieve ...

- ☐ ... a small training error.
- ☐ ... a large training error.
- ☐ ... a small test error.
- ☐ ... a large test error.
- ☐ ... a small generalization error.
- ☐ ... a large generalization error.

2. An overfitted model has ...

- ☐ ... a large test error.
- ☐ ... a small test error.
- ☐ ... a large training error.
- ☐ ... a small training error.

3. An underfitted model has ...

- ☐ ... a large test error.
- ☐ ... a small test error.
- ☐ ... a large training error.
- ☐ ... a small training error.

4. A model tends to overfit when ...

- ☐ ... the training set is small.
- ☐ ... the regularization term has little weight.
- ☐ ... the capacity is smaller than the complexity of the task.
- ☐ ... the test error is close to the Bayes error.
- ☐ ... the training error is smaller than the Bayes error.

5. To prevent overfitting one can ...

- ☐ ... use a smaller test set.
- ☐ ... use a larger test set.
- ☐ ... use a smaller training set.
- ☐ ... use a larger training set.
- ☐ ... reduce the capacity of the model.
- ☐ ... increase the capacity of the model.

6. To prevent underfitting one can ...

- ☐ ... use a smaller test set.
- ☐ ... use a larger test set.
- ☐ ... use a smaller training set.
- ☐ ... use a larger training set.
- ☐ ... reduce the capacity of the model.
- ☐ ... increase the capacity of the model.

7. The goal of regularization is to reduce ...

- ☐ ... the training error.
- ☐ ... the generalization error.
- ☐ ... the test error.
- ☐ ... the Bayes error.

8. Mark the correct statements and correct the wrong ones.

- ☐ The test set is used to estimate the generalization error.
- ☐ The training set is used to control the training.
- ☐ The validation set is used to learn the task.
- ☐ The training error typically underestimates the generalization error by a smaller amount than the validation error.
- ☐ The validation set is used to learn the hyperparameters.

9. Given is a set of samples $\{x^{(1)}, \dots, x^{(m)}\}$ that are independently and identically distributed according to a uniform distribution on the interval $[-0.8, 1.2]$.

(a) Check whether the sample mean

$$\hat{\mu}_m = \frac{1}{m} \sum_{i=1}^m x^{(i)} \quad (1.1)$$

is an unbiased estimator of the true mean μ .

(b) Assume that the absolute value of each sample is accidentally taken before the sample mean value is calculated. Thus the new estimator is

$$\hat{\mu}_m = \frac{1}{m} \sum_{i=1}^m |x^{(i)}|. \quad (1.2)$$

Determine the bias of this poor estimator to the mean μ of the initial distribution.

(c) How can the estimator of (b) be fixed so that he still gives an unbiased estimate?

10. **Optional:** Consider a set of samples $\{x^{(1)}, \dots, x^{(m)}\}$ that are independently and identically distributed according to a uniform distribution on the interval $[0, \theta]$, thus

$$p(x^{(i)}, \theta) = \begin{cases} \frac{1}{\theta}, & 0 \leq x^{(i)} \leq \theta \\ 0, & \text{otherwise} \end{cases}.$$

A biased estimator for the parameter θ is

$$\hat{\theta} = \max(x^{(1)}, \dots, x^{(m)}).$$

Correct this estimator so that it becomes an unbiased estimator for θ .